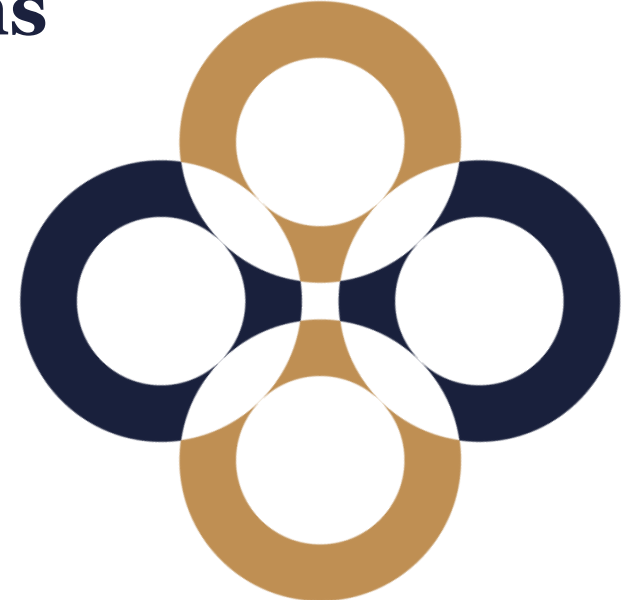


Credit risk I.

Risk Management and Financial Institutions

Zsuzsanna Vőneki



Credit risk

- Default risk: the probability that the borrower or our partner will not be able to pay back their debt
- Credit risk:
 - **Wider than default risk**
 - **The change in the debtor's default probability is also included**

Elements of credit risk

- The present value of the expected loss on credit risk depends on:
 - **PD = probability of default**
 - **Percentage of our claim what we lose in case of default (LGD = loss given default)**
 - **EAD = exposure at default**
 - **Yield used for discounting (PV ())**
- **$PV(\text{Expected loss}) = PD * LGD * EAD$**

Basel regulation

Standard method: risk weights specified by the regulator

IRB: Based on internal rating; the risk weighting function is determined by the regulator; internal estimate for PD;

Advanced IRB: the risk weighting function is determined by the regulator; Internal estimation for all parameters



IRB method

VAR:

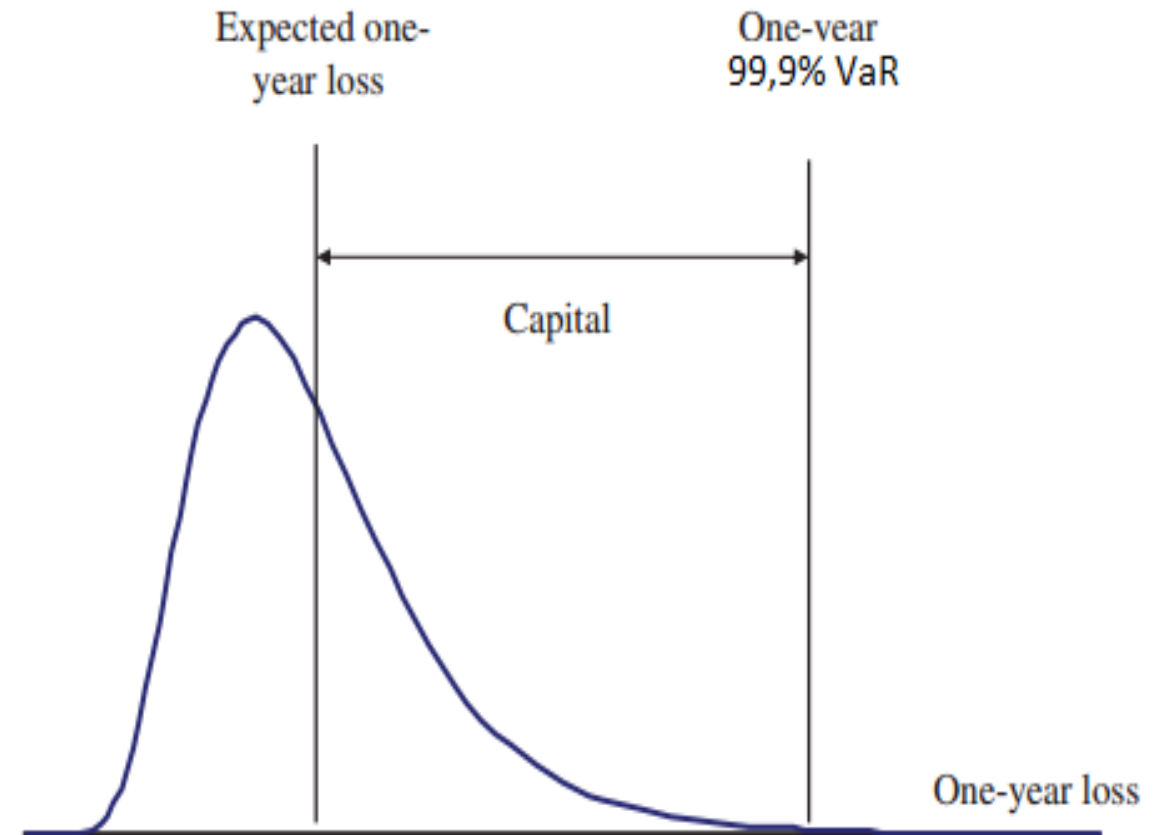
$$\sum_i EAD_i \cdot LGD_i \cdot WCDR_i$$

Expected loss:

$$\sum_i EAD_i \cdot LGD_i \cdot PD_i$$

Capital:

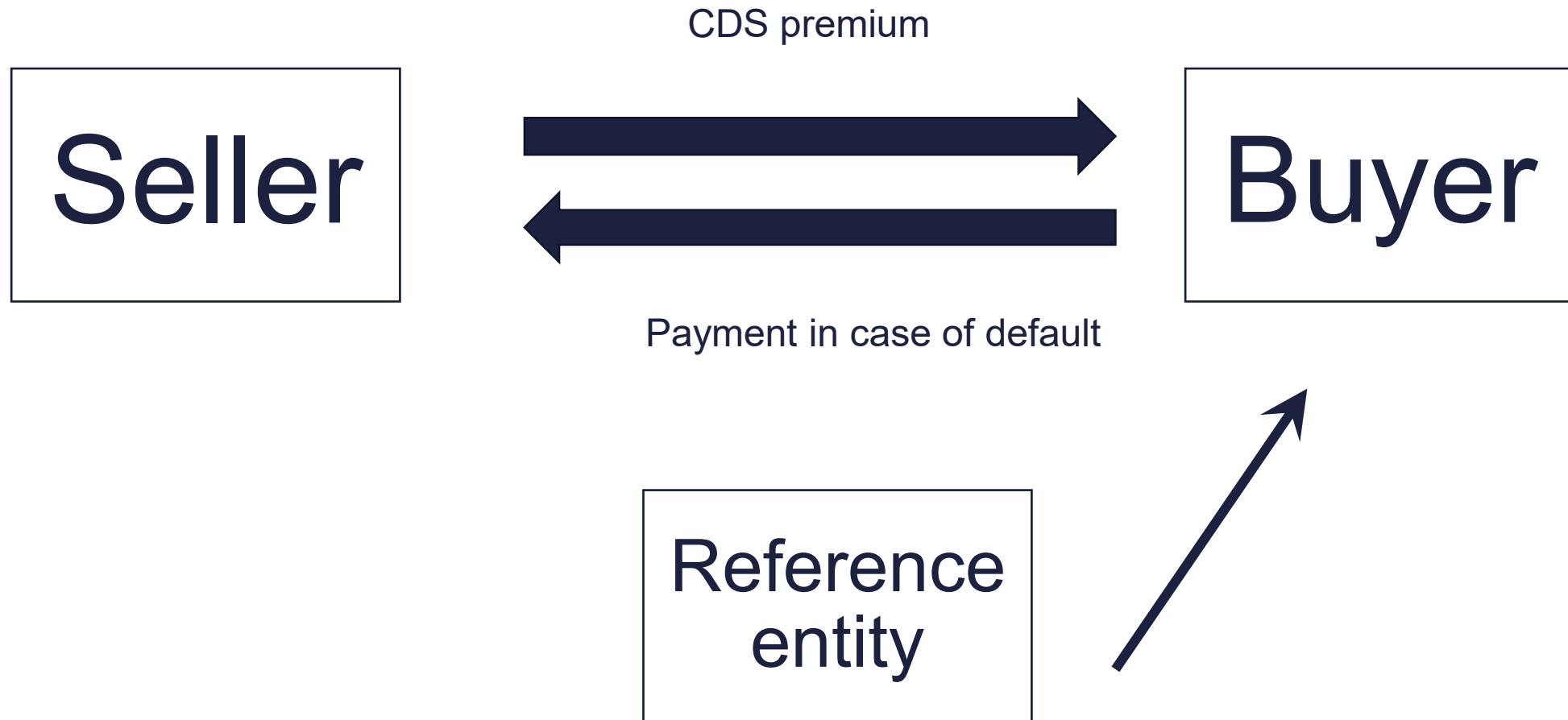
$$\sum_i EAD_i \cdot LGD_i \cdot (WCDR_i - PD_i)$$



PD estimation methods

- 1. From the yield curves of risky bonds (actuarial approach), e.g. Hull-White model
- 2. Structural models, e.g. Merton model, KMV model
- 3. Rating: a transition matrix from the past evolution of ratings, e.g. Creditmetrics

Credit Default Swap (CDS)



Average Cumulative Default Rates

Rating	Time Horizon (years)							
	1	2	3	4	5	7	10	15
AAA	0.00	0.03	0.13	0.24	0.34	0.51	0.70	0.90
AA	0.02	0.06	0.11	0.21	0.30	0.49	0.70	0.99
A	0.05	0.13	0.22	0.33	0.46	0.76	1.20	1.84
BBB	0.16	0.43	0.75	1.14	1.54	2.27	3.24	4.54
BB	0.63	1.93	3.46	4.99	6.43	8.89	11.64	14.65
B	3.34	7.80	11.75	14.89	17.35	20.99	24.62	28.24
CCC/C	28.30	38.33	43.42	46.36	48.58	50.75	52.76	54.76

CCC/C Bond 38,33% chance of defaulting for the end of the second year, 43,42% chance of defaulting for the end of the third year. Default during the third year: $43,42 - 38,33 = 5,09\%$ (unconditional default probability – as seen at time zero)

$100 - 38,33 = 61,67\%$ - Bond will survive until the end of year two

The probability that it will default during the third year conditional on no earlier default is: $0.0509 / 0.6167$ or 8.25% (conditional probability or hazard rate or default intensity)

Hazard Rate vs. Unconditional Default Probability

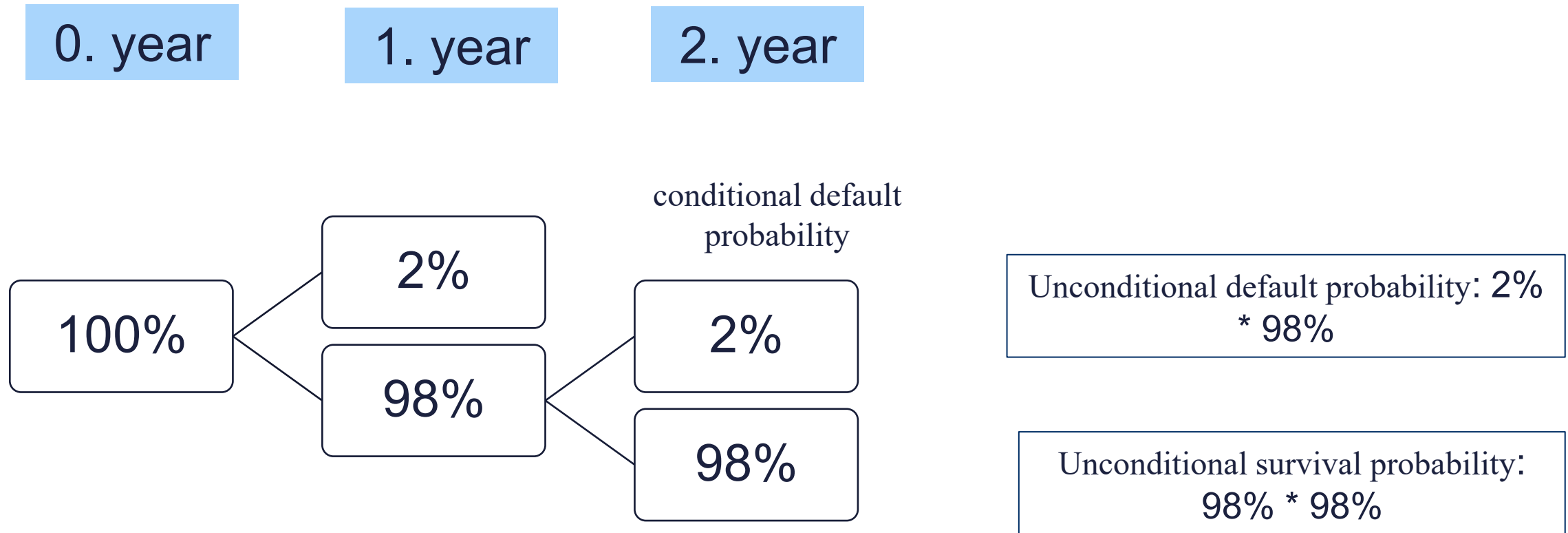
- The hazard rate or default intensity is the probability of default over a short period of time conditional on no earlier default .

$V(t) - V(t + \Delta t)/V(t) = \lambda(t)\Delta t$, $V(t)$ is the cumulative probability of the company surviving to time t

$Q(t) = 1 - e^{-\lambda(t)*t}$, where $\lambda(t)$ is the average hazard rate between time zero and time t ,
 $Q(t)$ probability of default by time t .

- The unconditional default probability is the probability of default as seen at time zero .

Conditional and unconditional default probability



Back test

Binomial distribution:

$$P(x = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Kupiec test:

$$-2 \ln[(1 - p)^{n-m} p^m] + 2 \ln[(1 - m/n)^{n-m} (m/n)^m]$$

Binomial: below 5% we reject the goodness of the model

Kupiec: above 3,84 we reject the goodness of the model



**Thank you
for your attention!**

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